Provenance of 'dirac16complex: a second-quantized Pin(4,4) fermion field and the 6-planes it creates'

1. Sources

- **Pre-gravity6plane.nb** (P. L. Nash, (C) 2022, GPL), read in full. Used: coordinates $X = \{x_0, \ldots, x_7\}$ (cell 45); $\eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$ (cell 66); σ (cell 73); τ_A, $\bar{\tau}_A = \sigma(\sigma\tau_A)^T$ (cell 265); $T^A = (T16^A)[A] = \begin{pmatrix} 0 & \bar{\tau}_A \\ \tau_A & 0 \end{pmatrix}$ (cell 308); $\sigma_{16} = T^0T^1T^2T^3$ (cells 306, 333); the Universe wave function $\Psi_{16} = f16[k][x_0, x_4]$ (cell 78); the Lagrangian $L_a = \frac{1}{H}\Psi_{16}^T\sigma_{16}\sum_A T^A\partial_A\Psi_{16} + \frac{2M}{H}\Psi_{16}^T\sigma_{16}\Psi_{16}$ (cell 1057); its Euler-Lagrange equations eLa and the coupling sets $\{0,5,8,13\}, \{1,4,9,12\}, \{2,7,10,15\}, \{3,6,11,14\}$ (cells 1072-1095); the variables $z = 6Hx_0$, $t = Hx_4$ (cell 61) and eLazt (cell 1099); the Maple exponential solutions with exponent $C(t + \frac{z}{6}) - \frac{M^2}{CH^2}(t - \frac{z}{6})$ (cells 1172-1193); $z_{adv} = 6t + z$, $z_{ret} = 6t - z$ and $\xi_{adv} = x_0 - x_4$, $\xi_{ret} = x_0 + x_4$ (cells 1203, 1246); vanishing bilinears of light-like quanta (cells 1208-1234); unit spinor $u = \frac{1}{\sqrt{2}}(1,0,0,0,1,0,0,0)$ (cells 783-801); frames ${F_A}^a$, ${F_a}^A$ (cells 888-893); vectorFormOfType1/Type2, the 7-planes $0 = V \cdot Z$, M^6 as their intersection and the count $\binom{8}{2} = 28$ (cells 1255-1294).
- **User instructions** (2026-10-30): define dirac16complex as a 16-component Pin(4,4) field with complex anticommuting Grassmann components, a second-quantized fermion field; use $(T16^A)[0..7]$ as Dirac matrices; add quadratic mass and quartic self-coupling terms (compare with Higgs couplings); mirror Pre-gravity6plane.nb and create 6-dimensional planes.
- **Standard physics** used: Grassmann (Berezin) calculus with left derivatives and order-reversing conjugation $(\theta\eta)^* = \eta^*\theta^*$; canonical quantization of fermions $\{\psi_a(x), \pi_b(y)\} = i\delta_{ab}\delta(x-y)$; Noether's theorem for $\psi \to e^{i\alpha}\psi$ giving $J^A = \bar{\psi}T^A\psi$; indefinite-metric (Krein space) quantization as used for example in Gupta-Bleuler quantization.

2. Computations (Wolfram Language 15.0.1, exact arithmetic, 2026-10-30)

All results quoted in the guide are outputs of 6planes.nb (generated from 6planes_src.txt by build_nb2.wls; 34 input cells, 0 messages):

result	6planes.nb
T^A real, $\{T^A, T^B\} = 2\eta^{AB}I_{16}$, σ_{16} and S^{AB} properties	Part 1, Out[3]-Out[6]
Grassmann engine checks	Part 2, Out[7]
$\mathcal{L}$ has 392 monomials, mass and quartic terms non-zero, $\mathcal{L}^* = \mathcal{L}$	Part 3.2, Out[8]
EL equations equal $\sigma_{16}(T^A\partial_A\psi - m\psi - \lambda(\bar{\psi}\psi)\psi)$; conjugate equation	Part 3.3, Out[9]; component 0 in Out[10]
$\partial_A J^A = 0$ on shell	Part 3.4, Out[11]
$K = -i\sigma_{16}T^4$ Hermitian, $K^2 = I$, spectrum $8\times(+1)$, $8\times(-1)$; $g_{45}^T\sigma_{16}T^4g_{45} = -\sigma_{16}T^4$; $\sigma_{16} = K\beta$	Part 3.5, Out[12]
$[K, T^A] = 0$ $(A \le 4)$, $\{K, T^A\} = 0$ $(A \ge 5)$, no cross terms, $\pm$ sign of the two sectors	Part 3.6, Out[13]
eLa-analogue for dirac16complex depending on x_0, x_4, and its couplings	Part 4.1-4.2, Out[14]-Out[15]
exponential (Maple-family) solutions and plane waves solve the equation	Part 4.3, Out[16]-Out[17]
light-like quanta: $J^A = 0$, $\bar{\psi}\psi = 0$	Part 4.4, Out[18]
$u^T\sigma u = 1$, ${F_A}^a{F_a}^A = I_8$	Part 4.5, Out[19]
M^6: rank 2, 28 valid parameterizations, explicit z_7, z_8	Part 4.6, Out[20]
pair 6-plane: rank 2	Part 4.6, Out[21]-Out[22]

The frames ${F_A}^a$, ${F_a}^A$ were also compared with the stored outputs (cells 889 and 891) of Pre-gravity6plane.nb: identical.

3. New in this work

The Grassmann-valued field, the order-reversing reality proof, the quartic term $-\frac{\lambda}{2}(\bar{\psi}\psi)^2$, the Krein-metric theorem with the explicit element $g_{45} = T^4T^5$, the mirror-sector decomposition $\mathcal{L} = \mathcal{L}_D[\psi_+] - \mathcal{L}_D[\psi_-] - \frac{\lambda}{2}(S_+ - S_-)^2$, the identification of the old Maple solutions as the light-cone exponential family $(6p+q)(6p-q) = m^2/H^2$, the definition of M^6 through the real part of a complex Universe, and the pair 6-plane M^6_{pair}.